main.py

Visualize

Run ▶

```python
N = 8
solutions = []

def safe(cols, row):
    c = len(cols)
    return all(r != row and abs(r - row) != c - i for i, r in enumerate(cols))

def place(cols):
    if len(cols) == N:
        solutions.append(cols[:])
        return
    for row in range(N):
        if safe(cols, row):
            cols.append(row)
            place(cols)
            cols.pop()

place([])
print("Total solutions:", len(solutions))
print("First solution (row index for each column):", solutions[0])
for r in range(N):
    print(' '.join('Q' if solutions[0][c] == r else '.' for c in range(N)))
```

Output

```
Total solutions: 92
First solution (row index for each column): [0, 4, 7, 5, 2, 6, 1, 3]
Q . . . . . . .
. . . . . . Q .
. . . . Q . . .
. . . . . . . Q
. Q . . . . . .
. . . Q . . . .
. . . . . Q . .
. . Q . . . . .
```